

Orlando Alexander

✉ orlando@orlandoalexander.uk ☎ +44 7486372351 📁 Project Portfolio 🌐 LinkedIn 🐙 GitHub

EDUCATION

BSc Data Science, *University of Bristol (2024 - 2027)*

- First-Class in Year 1: Top marks in Algorithms & Programming (87%) and Coding & Data Analysis (83%).
- Wrote a renewable energy paper (85%) using Python to analyse 28-year global dataset, revealing inverse GDP-fossil fuel consumption correlation and eight-fold renewable energy growth in high-income countries (2000-2015).
- Proficient in full-stack development (Python, JavaScript, React, Flask, SQL) and R, C/C++, AWS, GCP, and GitHub.

A-Levels, *Berkhamsted Sixth Form (2020 - 2022)*

- A*, A*, A*, A in Physics, Computer Science, Mathematics, and Further Mathematics.
- A* (100%) in EPQ: Designed a flow-optimised hydroelectric energy generator to maximise power output.
- Head Boy: Founded Academic Gala; delivered talks to 1,000+ audience members; campaigned on food waste.

PROFESSIONAL EXPERIENCE

Head of Software Engineering & Data Science, *180 Degrees Consulting, Bristol (Sep 2024 - Present)*

- Lead 36 consultants delivering pro-bono projects for 7+ social impact clients; achieved 100% satisfaction rates.
- Scaled division by 112% in six months (17 to 36 consultants), attracting over 450 applicants in 2025.
- Secured Le Wagon (#1 global tech bootcamp) partnership to train consultants in Software Engineering and AI.
 - **Data Pipeline for Disease Analysis**, *EcoSwell (Jun - Sep 2025)*
 - Implemented an end-to-end data pipeline consolidating 66,000 entries from 5 years of weather, dengue, census, and expenditure data using Pandas, NumPy, and Selenium, replacing weeks of manual downloading.
 - Used Monte Carlo analysis to forecast dengue outbreaks in Peru, enabling data-driven public health decisions.
 - **Resource Management Web App**, *everyFAMILY (Feb - Apr 2025)*
 - Deployed React data management web app with Flask REST API backend (Docker on GCP Cloud Run).
 - Reduced onboarding time by 30%; delivered scalable platform rated 5/5 by client for 'exceeding expectations'.
 - **Food Aid Visualisation Dashboard**, *Cirencester Foodbank (Sep - Dec 2024)*
 - Developed full-stack analytics dashboard with Streamlit frontend and Python backend (Pandas, NumPy, Matplotlib), implementing REST API to process and visualise 5,000+ food aid records across 100 data fields.
 - Delivered 3 years of insights on regional demand and seasonal trends to enable targeted aid allocation.

Full-Stack Software Engineer, *Educatch, London (May 2024 - Present)*

- Designed and deployed production-grade React platform with MySQL backend, automating scheduling, invoicing, and attendance analytics for 50+ students.
- Reduced administrative workload by 10 hours/week; became company's most-used tool, supporting expansion.

Junior Software Engineer, *WISENSE, Berlin (Feb 2023 - Feb 2024)*

- Built ML studio with drag-and-drop interface to create video analysis workflows in <5 mins using React and Axios.
- Deployed for major clients including BMW, ABB, and SIG for real-time manufacturing anomaly detection.

Junior Software & AI Engineer, *Connect AI (Jan 2023 - Jan 2024)*

- Developed GPT-4 LLM-powered customer support assistant with React frontend for startups including Digital Republic, handling 5,000 messages/month with 24/7 multilingual support.
- Implemented RAG workflows with real-time feedback loops, reducing first-level ticket volume by 20% and improving resolution time by 95%.

Renewable Energy Intern, *EcoSwell, Peru (Aug - Sep 2022)*

- Developed integrated monitoring system with Python on Raspberry Pi, capturing 10,000+ data points weekly across 8 key data types.
- Designed offline logging system for automated, reliable reporting on UN SDG goals in remote conditions.

Computer Vision Engineer Intern, *Inovo Robotics, Guildford (Aug 2021)*

- Designed computer vision system for automated pastry decoration, 90% more efficient than manual methods.
- Implemented advanced algorithms including Canny edge detection, Hough transformations, and clustering for path optimisation using OpenCV and mlrose.

Technical Project Lead, *Young Enterprise, Berkhamsted (Sep 2020 - Apr 2021)*

- Built company website processing 300+ sales, generating £6,000 revenue and receiving digital innovation award.
- Led company operations using Excel and Trello, earning Young Enterprise Journey Award for technical leadership.

INTERESTS

Programming Tutor — Tutored 300+ students in Python, Web Development and ML at PMT, codetoday and Code Kids.

Theatre — Performed at Edinburgh Fringe 2024; produced and acted in 15 Bristol DramSoc productions.